The Ontic Mathematical Foundations: From Counting to Coupling

The Constant Chain $\gamma \to \varpi \to M \to \pi \to G^*$

William J. Steinmetz III[1]

[1] *Foundational Ternary Dynamics Research*

February 14, 2026

Framework: FTD v5.24

Abstract

We formalize the ontic foundation of mathematical constants in Foundational Ternary Dynamics (FTD) as a directed chain of emergence: $\gamma \to \varpi \to M \to \pi \to G^*$, where γ is the Euler–Mascheroni constant, ϖ is the lemniscate constant, $M = \mathrm{AGM}(1, \sqrt{2})$ is the Gauss arithmetic-geometric mean, π is Archimedes' constant, and G^* is the lemniscate-alpha constant of FTD. The chain is grounded by a rigorous structural claim: γ **is the inversion term where the discrete integer world first makes contact with the continuous analytical world, and this contact is logarithmic in character.** Without γ, there is no bridge from discrete counting to the elliptic and lemniscatic structures from which the fine-structure constant (α) emerges. The ordering is defined not by numerical value, but by ontological depth.

Epistemic Tags: *Throughout this document, we utilize strict epistemic bounding:* [**Theorem**] *for structural results,* [**Definition**] *for defining relations,* [**Standard**] *for classical analysis,* [**Selection**] *for ordering, and* [**Conjecture**] *for physical interpretation.*

1. The Five Constants

1.1 The Euler–Mascheroni Constant γ

$$\gamma = \lim_{n\to\infty} \left(\sum_{k=1}^{n} \frac{1}{k} - \ln n \right) = 0.5772156649\ldots \tag{1}$$

What it requires: Only the positive integers and the concept of a limit.
What it encodes: The exact offset between *discrete counting* (the harmonic series $H_n = 1 + 1/2 + 1/3 + \cdots + 1/n$) and *continuous scaling* (the natural logarithm $\ln n$). This makes γ the **universal conversion constant** between the arithmetic world of integers and the analytic world of continuous functions.
The inversion property: Rearranging $H_n \approx \ln(n) + \gamma$, we find

$$\exp(H_n - \gamma) \approx n \tag{2}$$

Without γ, one cannot invert from continuous logarithmic scaling back to discrete counting. It is the constant that renders the integers recoverable from their logarithmic shadow.
Why γ is ontologically first: Every subsequent constant in the chain depends on the Gamma function $\Gamma(z)$, and γ appears in its Weierstrass product representation:

$$\frac{1}{\Gamma(z)} = z \cdot e^{\gamma z} \cdot \prod_{n=1}^{\infty} \left[\left(1 + \frac{z}{n}\right) e^{-z/n}\right] \tag{3}$$

Here γ is the **exponential rate** that controls how the Gamma function scales. It is embedded multiplicatively—as $\exp(\gamma z)$—into every evaluation of Γ, including the critical value $\Gamma(1/4)$ from which ϖ and G^* are built.

1.2 The Lemniscate Constant ϖ

$$\varpi = \frac{\Gamma(1/4)^2}{2\sqrt{2\pi}} = 2.6220575543\ldots \tag{4}$$

What it requires: γ (through $\Gamma(1/4)$), the integer 4, and π.
What it encodes: The half-period of the Bernoulli lemniscate $r^2 = \cos(2\theta)$—the "π of the lemniscate." Just as π measures the circle, ϖ measures the figure-eight: the first self-crossing algebraic curve, encoding self-reference geometrically.
Connection to γ: The digamma function satisfies

$$\psi(1/4) = -\gamma - \frac{\pi}{2} - 3\ln 2 \tag{5}$$

This identity shows that γ is *analytically present* in the logarithmic derivative of Γ at the critical point $z = 1/4$. Since $\Gamma(1/4)$ determines ϖ, γ flows through to ϖ via the derivative chain:

$$\gamma \xrightarrow{\text{Weierstrass}} \Gamma(1/4) \xrightarrow{\text{definition}} \varpi \tag{6}$$

1.3 The Gauss Constant M

$$M = \mathrm{AGM}(1, \sqrt{2}) = 1.1981402347\ldots \tag{7}$$

What it requires: The arithmetic-geometric mean iteration applied to 1 and $\sqrt{2}$.
What it encodes: The rate of convergence of the AGM process—itself a bridge between arithmetic means (discrete averaging) and geometric means (multiplicative scaling). The AGM is the fastest-converging classical algorithm, doubling precision with each step.

[Theorem] Exact relationship:

$$\varpi = \frac{\pi}{M} \qquad \Longleftrightarrow \qquad M = \frac{\pi}{\varpi} \tag{8}$$

This identity reveals M as the **conversion factor between circular geometry (π) and lemniscatic geometry (ϖ)**. The reciprocal $1/M \approx 0.8346$ is the classical Gauss constant.
Note on ordering: M sits between ϖ and π in the chain because it is the bridge that **converts** between them. ϖ is more fundamental (requiring only $\Gamma(1/4)$ and hence only γ and 4); M requires the AGM concept; π requires circular geometry or, equivalently, the combination $M \cdot \varpi$.

1.4 Archimedes' Constant π

$$\pi = M \cdot \varpi = \frac{4\varpi^2}{G^{*2}} = 3.1415926536\ldots \tag{9}$$

What it requires: The circle, or equivalently, the product of ϖ and M.
What it encodes: The ratio of circumference to diameter—the constant of **circular closure**. In FTD, π is not ontologically primitive. It is the product of two more fundamental constants:

$$\pi = \varpi \cdot M = \varpi \cdot \mathrm{AGM}(1, \sqrt{2}) \tag{10}$$

This decomposition reveals π as a **composite**: the lemniscatic half-period (ϖ) scaled by the arithmetic-geometric bridge (M).

[**Selection**] The claim that π is ontologically derivative of ϖ is a selection principle. Mathematically, any of $\{\gamma, \varpi, M, \pi, G^*\}$ can be expressed in terms of the others. The ordering reflects the principle that constants requiring less structure are more fundamental.

1.5 The Lemniscate-Alpha Constant G^*

$$G^* = \frac{\sqrt{2}\,\Gamma(1/4)^2}{2\pi} = \frac{2\varpi}{\sqrt{\pi}} = 2.9586751192\ldots \tag{11}$$

What it requires: All four preceding constants, or equivalently, ϖ and π together.
What it encodes: The master coefficient of the FTD quadratic [1]:

$$x^2 - 16\,G^{*2}\,x + 16\,G^{*3} = 0 \tag{12}$$

whose roots are $x_+ = 137.036\ldots \approx 1/\alpha$ and $x_- = 3.024\ldots \approx N_c$.
Why G^* is ontologically last: It is the most "constructed" of the five. It requires $\sqrt{2}$ (from the Gauss constraint geometry), $\Gamma(1/4)^2$ (which carries γ through Weierstrass), and π (which carries M and ϖ). G^* is where all the preceding structure **converges** into a single number that, through the master quadratic, produces physics.

2. The Chain as Ontological Ordering

2.1 The Principle

The ordering $\gamma \to \varpi \to M \to \pi \to G^*$ is not by numerical magnitude but by **ontological depth**: the amount of mathematical structure required for each constant's definition.

Level	Constant	Value	Requires	Encodes
0	γ	0.5772...	Integers + limit	Discrete $\leftrightarrow$ continuous bridge
1	ϖ	2.6221...	γ (via Γ), integer 4	Self-crossing geometry
2	M	1.1981...	AGM iteration	Arithmetic $\leftrightarrow$ geometric bridge
3	π	3.1416...	$\varpi \times M$	Circular closure
4	G^*	2.9587...	$\sqrt{2}$, ϖ, π	Master quadratic $\to \alpha$

Table 1: The ontic hierarchy of mathematical constants.

2.2 Each Level Adds Structure

Level 0 → 1 ($\gamma \to \varpi$): From discrete/continuous bridging to self-crossing geometry. The Weierstrass product converts γ (a limit of sums) into $\Gamma(1/4)$ (a special function value), and the definition $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi})$ crystallizes this into the period of a self-referential curve. This step adds **geometric self-reference**.

Level 1 → 2 ($\varpi \to M$): From lemniscatic period to the AGM convergence rate. $M = \pi/\varpi$ is defined by the arithmetic-geometric mean, which iterates between two types of averaging. This step adds **iterative convergence**.

Level 2 → 3 ($M \to \pi$): From AGM rate to circular constant. $\pi = \varpi \cdot M$ combines the self-crossing period with the convergence bridge. This step adds **closure** (the circle is the simplest closed curve).

Level 3 → 4 ($\pi \to G^*$): From circular to lemniscatic with $\sqrt{2}$ scaling. $G^* = 2\varpi/\sqrt{\pi} = \sqrt{2}\cdot\Gamma(1/4)^2/(2\pi)$ introduces the $\sqrt{2}$ from the FCC/Gauss constraint geometry. This step adds **constraint physics**—the discrete lattice's Gauss law.

2.3 γ as the Logarithmic Inversion Term

The deepest claim: γ **is where numbers first scale logarithmically into the decimal space.**

The harmonic series $H_n = 1 + 1/2 + \cdots + 1/n$ grows as $\ln(n) + \gamma$. This means the "natural" way to accumulate the reciprocals of integers (a purely arithmetic operation on $1, 2, 3, \ldots$) produces a logarithmic function plus a correction term γ. The integers, through their reciprocal sum, *generate* logarithmic scaling—and γ is the remainder.

More precisely:

- **Discrete world:** $H_n = \sum 1/k$ (sum over integers)
- **Continuous world:** $\ln(n)$ (integral $\int dx/x$)
- **Bridge:** $\gamma = H_n - \ln(n)$ in the limit

The exponential form $e^{-\gamma} \approx 0.5615$ acts as a **decimal discount factor**: it governs how the discrete harmonic world undershoots the continuous logarithmic world. By Mertens' theorem [3], $e^{-\gamma}$ even controls the distribution of primes:

$$\prod_{p \le N,\, p \text{ prime}} \left(1 - \frac{1}{p}\right) \sim \frac{e^{-\gamma}}{\ln N} \tag{13}$$

So γ is not merely a correction term—it is the constant that governs how **prime factorization** (the most fundamental discrete structure) maps to **logarithmic density** (the most fundamental continuous measure).

3. The Complete Derivation Chain

3.1 From γ to α

The full chain, with each step justified:

$$\boxed{\gamma \xrightarrow{\text{Weierstrass}} \Gamma(1/4) \xrightarrow{\text{def.}} \varpi \xrightarrow{\text{AGM}} M \xrightarrow{\times\varpi} \pi \xrightarrow{\sqrt{2}\,\text{scaling}} G^* \xrightarrow{\text{quadratic}} \alpha} \tag{14}$$

1. $\gamma = 0.5772\ldots$ (integers + limit)
2. Weierstrass product: $1/\Gamma(z) = z \cdot e^{\gamma z} \cdot \prod \ldots$; evaluate at $z = 1/4$
3. $\Gamma(1/4) = 3.6256\ldots$
4. Definition: $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi}) = 2.6221\ldots$
5. AGM identity: $M = \pi/\varpi = \mathrm{AGM}(1, \sqrt{2}) = 1.1981\ldots$
6. Product: $\pi = \varpi \cdot M = 3.1416\ldots$
7. Scaling: $G^* = 2\varpi/\sqrt{\pi} = \sqrt{2} \cdot \Gamma(1/4)^2/(2\pi) = 2.9587\ldots$
8. Master quadratic: $x^2 - 16G^{*2}x + 16G^{*3} = 0$
9. $x_+ = 137.036\ldots \approx 1/\alpha, \quad x_- = 3.024\ldots \approx N_c$

3.2 The Defining Relations

[**Definition**] The entire chain rests on exactly **two independent definitions** plus one **FTD-specific claim**:

Definition 3.1 (Lemniscate Constant)**.** $\varpi = \dfrac{\Gamma(1/4)^2}{2\sqrt{2\pi}}$

Definition 3.2 (Lemniscate-Alpha Constant)**.** $G^* = \dfrac{\sqrt{2}\,\Gamma(1/4)^2}{2\pi} = \dfrac{2\varpi}{\sqrt{\pi}}$

[**Conjecture**] The master quadratic with coefficient 16 (from lattice degrees of freedom):

$$x^2 - 16\,G^{*2}\,x + 16\,G^{*3} = 0 \tag{15}$$

has roots $x_+ = 137.036\ldots \approx 1/\alpha$ and $x_- = 3.024\ldots \approx N_c$.

3.3 The γ–ϖ Connection: How Counting Becomes Geometry

[**Theorem**] This is the genuinely meaningful part of the chain—the mechanism by which the discrete-to-continuous bridge constant γ flows into the lemniscatic structure.

The **Weierstrass product** provides the link:

$$\frac{1}{\Gamma(z)} = z \cdot e^{\gamma z} \cdot \prod_{n=1}^{\infty} \left[\left(1 + \frac{z}{n}\right) e^{-z/n}\right] \tag{16}$$

Evaluated at $z = 1/4$ (the FTD-selected evaluation point), γ enters as the factor $\exp(\gamma/4)$ inside $\Gamma(1/4)$. Since ϖ and G^* depend on $\Gamma(1/4)^2$, γ propagates through as $\exp(\gamma/2)$. This is not a numerical coincidence—it is a **structural consequence** of how the Gamma function is built from the integers.

The digamma function (logarithmic derivative of Γ) makes this connection explicit:

$$\psi(1/4) = -\gamma - \frac{\pi}{2} - 3\ln 2 \qquad \text{[\textbf{Standard}]—Gauss digamma theorem} \tag{17}$$

$$\psi(1/3) = -\gamma - \frac{3}{2}\ln 3 - \frac{\pi}{2\sqrt{3}} \qquad \text{[\textbf{Standard}]—Gauss digamma theorem} \tag{18}$$

These are classical identities due to Gauss (1813) [2]. What is structurally meaningful for FTD is that the evaluation points $z = 1/3$ and $z = 1/4$ correspond to the two primary framework integers $N_c = 3$ and $N_{\text{base}} = 4$. Setting both expressions equal (since both define γ) yields:

$$\psi(1/4) - \psi(1/3) = \frac{3}{2}\ln 3 + \frac{\pi}{2\sqrt{3}} - \frac{\pi}{2} - 3\ln 2 \tag{19}$$

This is a non-trivial **cross-constraint** linking the digamma at the two FTD integers through a specific combination of π, $\ln 2$, $\ln 3$, and $\sqrt{3}$.

3.4 Standard Results Used

[Standard] The chain uses several classical results from analysis. These are **tools**, not FTD discoveries:

Result	**Source**	**Role in Chain**
$\Gamma(z) \cdot \Gamma(1-z) = \pi / \sin(\pi z)$	Euler reflection	Relates $\Gamma(1/4)$ to $\Gamma(3/4)$
Gauss multiplication formula	Gauss (1812)	Connects Γ at multiples of $1/n$
$\psi(1/n)$ closed forms	Gauss (1813)	Links γ to $\Gamma(1/4)$ analytically
$\varpi \cdot M = \pi$	AGM–elliptic identity	Defines M as bridge
$\zeta(2n) = (-1)^{n+1} B_{2n} (2\pi)^{2n} / (2 \cdot (2n)!)$	Euler (1735)	Even zeta from Bernoulli numbers
$\prod(1 - 1/p) \sim e^{-\gamma} / \ln N$	Mertens (1874)	γ governs prime distribution

Table 2: Classical results employed in the derivation chain.

4. Structural Theorems

4.1 The Minimal Generating Set

Theorem 4.1 (Minimal Generating Set)**.** *The entire ontic chain is generated by exactly* ***two constants****: γ (Euler–Mascheroni) and π (Archimedes).*

Proof sketch. Given γ and π:

1. $\Gamma(1/4)$ is determined by the Weierstrass product (requiring only γ and the integer 4)
2. $\varpi = \Gamma(1/4)^2 / (2\sqrt{2\pi})$
3. $M = \pi/\varpi = \mathrm{AGM}(1, \sqrt{2})$
4. $G^* = 2\varpi/\sqrt{\pi} = \sqrt{2} \cdot \Gamma(1/4)^2/(2\pi)$
5. x_+, x_- from the master quadratic
6. $\alpha = 1/x_+$ (with precision corrections using framework integers)

The circularity (π appears in step 2, which seems to require π before generating it) resolves because **both γ and π are independently definable from the integers alone**: $\gamma = \lim[H_n - \ln(n)]$ and $\pi = 4\arctan(1) = 4\sum(-1)^n/(2n+1)$.

Implication: The fine structure constant α is determined by exactly two transcendental constants from number theory, plus the framework integers $\{3, 4, 7, 13\}$.

4.2 The $\exp(\gamma/2)$ Universal Scaling

Theorem 4.2 (Universal Rescaling)**.** *The factor $\exp(\gamma/2)$ is the* ***universal rescaling*** *that removes γ from all lemniscatic constants simultaneously.*

Define the "γ-free" constants:

- $\Gamma_0(1/4) = \Gamma(1/4) \cdot \exp(\gamma/4)$
- $\varpi_0 = \varpi \cdot \exp(\gamma/2) = \Gamma_0(1/4)^2/(2\sqrt{2\pi})$

- $G_0^* = G^* \cdot \exp(\gamma/2)$

Why $\exp(\gamma/2)$**?** Because γ enters the chain **only** through $\Gamma(1/4)$ via the Weierstrass product, where it contributes a factor of $\exp(\gamma/4)$. Since ϖ and G^* depend on $\Gamma(1/4)^2$, the γ-dependence always enters as $\exp(\gamma/4)^2 = \exp(\gamma/2)$.

Physical reading: The factor $\exp(-\gamma/2) \approx 0.749$ is the "discount" that the discrete-to-continuous bridge (γ) applies to the idealized lemniscatic constants. Without this discount, $\varpi_0 \approx 3.499$ and $G_0^* \approx 3.949$—both larger than their actual values. The integers, through their logarithmic shadow, **compress** the lemniscatic constants.

5. Relation to the FTD Hierarchy

5.1 Where This Fits

The existing FTD ontological hierarchy begins at Level 0 with the Pure Integral I_4. The ontic constant chain provides the **sub-structure beneath** I_4:

Prior Hierarchy	**This Document**	**Constant**
Level −3: Absolute Void	—	(no mathematics)
Level −2: Pregnant Void	—	(potentiality)
Level −1: First Distinction	Level 0	γ (integers emerge)
Level 0: Self-Reference ($n = 4$)	Level 0→1	$\gamma \to \Gamma(1/4)$
Level 1: Pure Integral I_4	Level 1	$I_4 = \varpi/2$
Level 5: Lemniscatic	Levels 1–4	$\varpi \to M \to \pi \to G^*$

Table 3: Integration of the ontic chain with the FTD hierarchy.

Key identification: $I_4 = \varpi/2$. The "Pure Integral" of the prior hierarchy is exactly half the lemniscate constant. The ontic chain $\gamma \to \varpi$ provides the foundation *beneath* I_4 by showing where the Gamma function (and hence ϖ) gets its scaling from.

5.2 The Role of the Integer 4

In the prior hierarchy, $n = 4$ is selected by self-reference. In the ontic chain, 4 appears at the **evaluation point** $z = 1/4$ where the Weierstrass product (carrying γ) is evaluated to produce $\Gamma(1/4)$. The two hierarchies converge:

- **Prior:** Self-reference selects $n = 4 \to I_4 = \int_0^1 dx/\sqrt{1 - x^4}$
- **Ontic:** γ flows through $\Gamma(z)$ at $z = 1/4 \to \Gamma(1/4) \to \varpi = 2I_4$

Both paths arrive at the same destination: the lemniscate constant ϖ. The integer 4 is the **meeting point** of the self-referential argument (why $n = 4$) and the analytic argument (why $z = 1/4$ in the Weierstrass product).

6. Summary and Physical Interpretation

6.1 The Ontic Chain

$$\boxed{\gamma \xrightarrow{\text{Weierstrass}} \Gamma(1/4) \xrightarrow{\text{def.}} \varpi \xrightarrow{\text{AGM}} M \xrightarrow{\times\,\varpi} \pi \xrightarrow{\sqrt{2}\,\text{scaling}} G^* \xrightarrow{\text{quadratic}} \alpha} \tag{20}$$

6.2 Physical Interpretation

[Conjecture] γ is the **first constant that exists** once the integers exist and the concept of a limit is available. It encodes the irreducible gap between counting and measuring—between the discrete and the continuous. Every subsequent constant in the chain inherits this gap through the Gamma function's Weierstrass product, which carries γ as an exponential rate factor.

The physical implication: **the fine structure constant α inherits its specific numerical value from γ through a chain of exact algebraic relationships.** The integers provide the skeleton ($\{3, 4, 7, 13\}$); γ provides the flesh (the continuous scaling); and the lemniscatic structure provides the self-referential closure that binds them together.

6.3 Epistemic Status

Claim	**Tag**	**Notes**
γ as logarithmic inversion term	**[Theorem]**	Follows from $H_n = \ln(n) + \gamma + O(1/n)$
Weierstrass: γ flows into $\Gamma(1/4)$	**[Theorem]**	Standard analysis; structural mechanism
$\exp(\gamma/2)$ universal scaling	**[Theorem]**	Consequence of $\Gamma(1/4)^2$ dependence
Minimal generating set $\{\gamma, \pi\}$	**[Theorem]**	Constructive proof
Digamma cross-constraint	**[Standard]**	Gauss (1813); FTD integers have closed forms
Ordering by ontological depth	**[Selection]**	Principled but not uniquely determined
Master quadratic yields $x_+ \approx 1/\alpha$	**[Conjecture]**	1.26 ppm match; no mechanism derived
Coefficient 16 from lattice DoF	**[Selection]**	Argued from geometry; product rule imposed

Table 4: Epistemic classification of all claims.

References

[1] W. J. Steinmetz III, "Gauge Couplings from Discrete Spacetime Geometry," *Foundational Ternary Dynamics Research*, Zenodo (January 2026). https://zenodo.org/records/18227181

[2] C. F. Gauss, "Disquisitiones generales circa seriem infinitam," *Commentationes societatis regiae scientiarum Gottingensis recentiores*, **2**, 1–46 (1813). *[Closed-form evaluations of the digamma function at rational arguments.]*

[3] F. Mertens, "Ein Beitrag zur analytischen Zahlentheorie," *Journal für die reine und angewandte Mathematik*, **78**, 46–62 (1874). *[Asymptotic product over primes: $\prod_{p \le N}(1-1/p) \sim e^{-\gamma}/\ln N$.]*

[4] L. Euler, “De summis serierum reciprocarum,” *Commentarii academiae scientiarum Petropolitanae*, **7**, 123–134 (1735). *[Solution to the Basel problem and evaluation of* $\zeta(2n)$ *via Bernoulli numbers.]*